The Great MTR Pivot, 1983-2019

John Green[1] Emma Kalish[2] Robert Moffitt[1]

[1]Johns Hopkins Economics

[2]Consumer Financial Protection Bureau

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Introduction

- Many changes to US social welfare programs over the last 40 years:
  - Cash welfare: Aid to Families with Dependent Children (AFDC) transitioned to Temporary Assistance for Needy Families (TANF)
  - Significant expansion of Earned Income Tax Credit (EITC)
  - Medicaid expanded for children from 1980s-2000s and Medicaid for adults expanded in the 2000s (depending on state)
  - Expansions to SNAP eligibility in 2000s

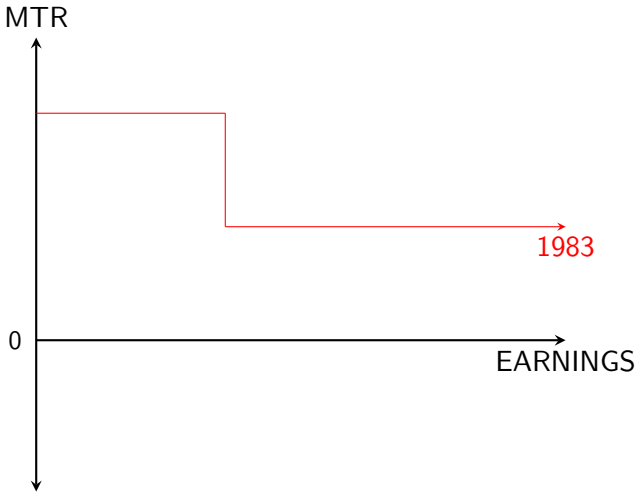
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- Net effect has been a significant expansion in benefits for workers
  - Emergence of the "work-based safety net"
- Result is the *Great MTR Pivot*: marginal tax rates (MTRs) have *fallen* at lower earnings points, but have (by definition) *increased* at higher earnings points
  - Before, MTRs could be near or even above 100% ("poverty traps") at low levels of earnings; now, are closer to (or even below!) 0
  - Ambiguous net effect on overall labor supply

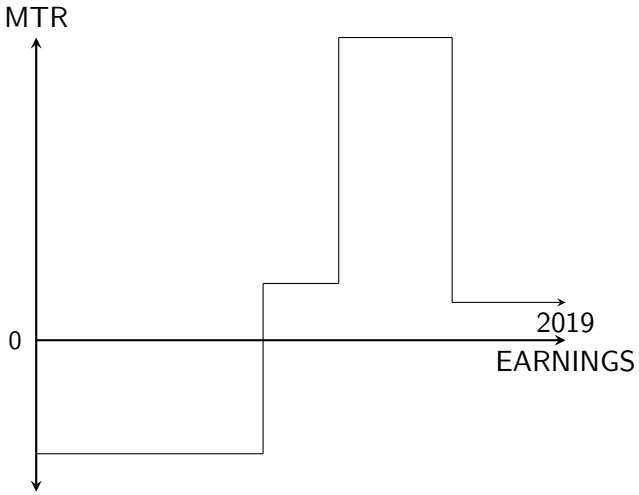
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  - Ambiguous net effect on overall labor supply
- Many programs are oriented towards **single mothers**, and we will focus on this group

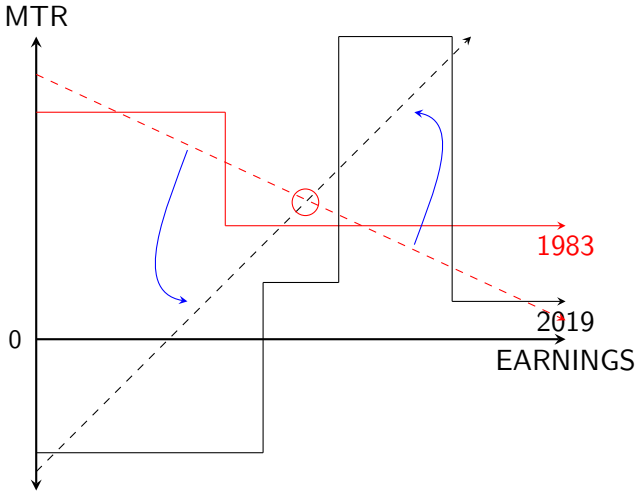
Stylized depiction



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This paper

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- **Second**: calibrate a model of program participation and labor supply of single mothers
  - Model a full set of social welfare programs
  - Match estimates from the literature of each program's labor supply effects
- **Third**: use the model to explore impact of actual and alternative rule changes on labor supply, consumption, and welfare

Prior work

- Negative income tax: Friedman [1961]
- Ambiguous welfare effects of EITC: Hansen [2021], Bierbrauer et al. [2023]
- Decrease in MTRs: Coe et al. [1998], CBO (2005, 2012, 2015), Maag et al. [2012], Kosar and Moffitt [2017], Chien and Macartney [2019], Hembre [2020]
- Incomplete takeup: Currie [2006], Ko and Moffitt [2024]
- LS models: Blundell et al. [2016], Guner et al. [2023]
- Heterogenous LS effects of different programs: various papers discussed below

Preview of results

Focus today on changes from **2000 to 2019** (skipping over AFDC to TANF transition)

- From 2000 to 2019 benefits *expanded*, especially at PT work
- Expansion driven by Medicaid and EITC, counteracted by decreases in TANF
- MTRs *fell* at the 0-to-PT margin, constant or *increased* at the PT-to-FT margin
- Changes to programs increased the employment rate (ER) by about 5 pp (6.5%)
- Consumption increases by about $290 monthly (12%) and utility by about 6.4%
- Atkinson measure for inequality *increases* for consumption, *decreases* for utility
- A simple revenue-neutral reform which raises baseline benefits (instead of lowering MTRs) would have generated significantly smaller welfare gains

Today

1. Introduction
2. Programs
3. Model
4. Data
5. Results
6. Reform
7. Conclusion

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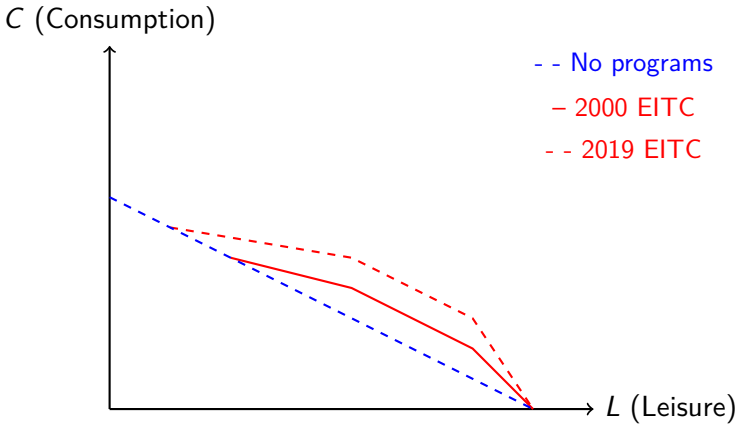
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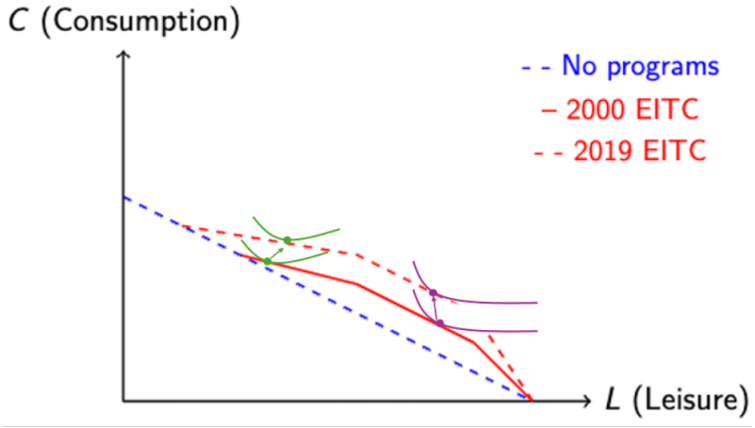
EITC

- Negative income tax which subsidizes work at low earnings levels
- Initially increases with income, flattens out, and then phases out
- Benefit increases with dependent children
- Many increases over the years at both federal (1986, 1990-1993, 2009) and state levels

EITC



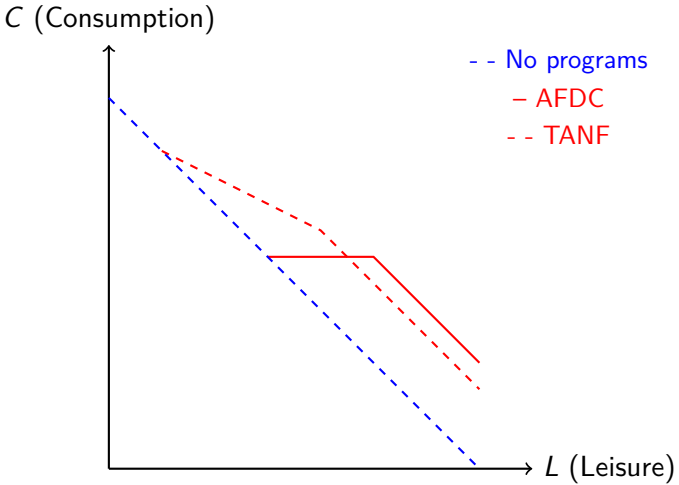
EITC with indifference curves



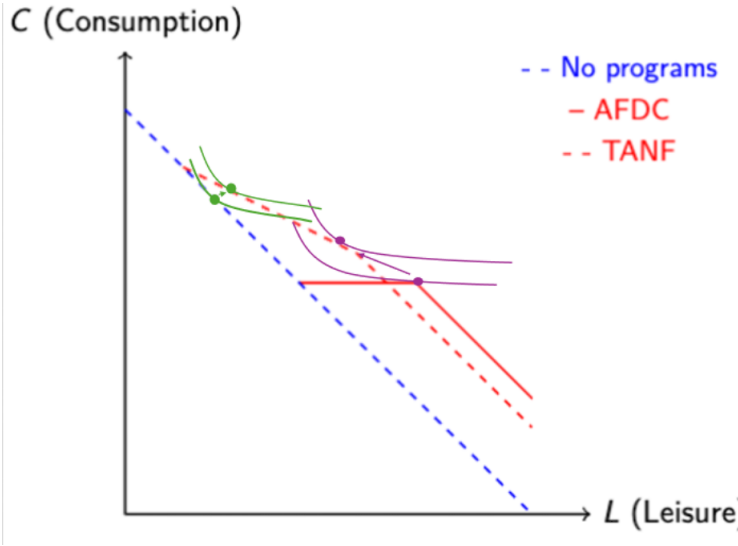
Cash welfare (AFDC and TANF)

- Provides cash payments to low-income families, largely those headed by single mothers
- Benefits increase with number of children but decrease with labor income
- In mid-1990s, Clinton led transition from AFDC to TANF. TANF is substantially different:
  - Under AFDC, benefits decreased approximately dollar-for-dollar; TANF lessened the work penalty
  - TANF benefits much harder to qualify for
  - Real value of benefits has eroded significantly
- Overall, cash welfare has gone from one of the most popular social welfare programs to a rarely used benefit

Cash welfare



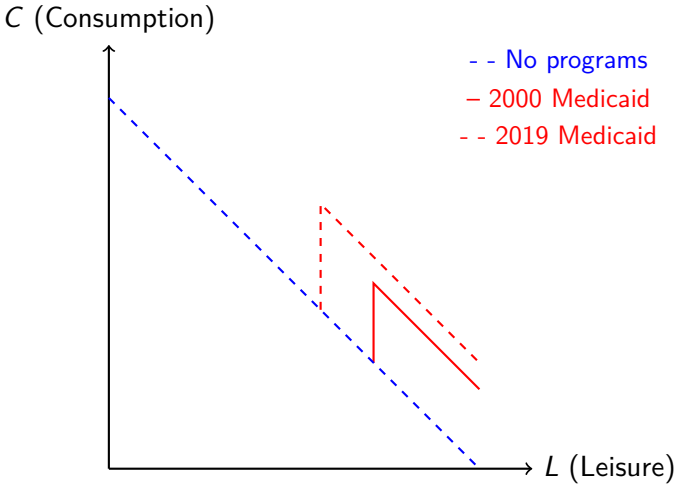
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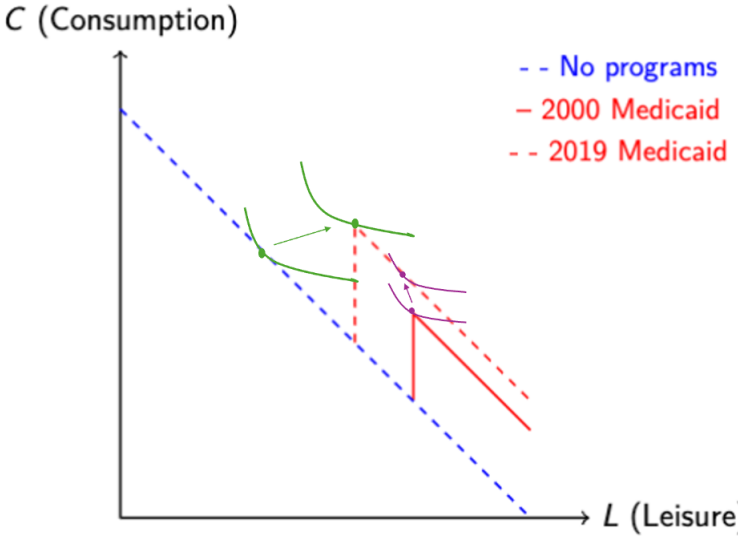
Medicaid

- Provides health insurance to low income individuals and families, either through classic Medicaid or through the Children's Health Insurance Program (CHIP)
- Originally aimed at children, and still easier for children to qualify: in many families, children but not adults are eligible
  - However, it has become increasingly available for adults and to those at higher income points, most notably through the ACA
- Classic "notch" program; one is either eligible for benefits or not depending on income
  - Federal and state reforms have moved the notch further out, but not changed the fundamental nature of the program

Medicaid


C (Consumption)
- - No programs
– 2000 Medicaid
- - 2019 Medicaid
L (Leisure)

Medicaid with indifference curves



Other programs

- Housing: program which subsidizes housing costs by covering a portion of median area rent which depends on income
  - Through the 80s and 90s, major transition from public housing to section 8 vouchers
  - In theory a high dollar in-kind transfer which has a large effect on MTRs; but in practice a *rationed* program which is not offered to all eligible people
  - Decreases smoothly with income before hitting a notch

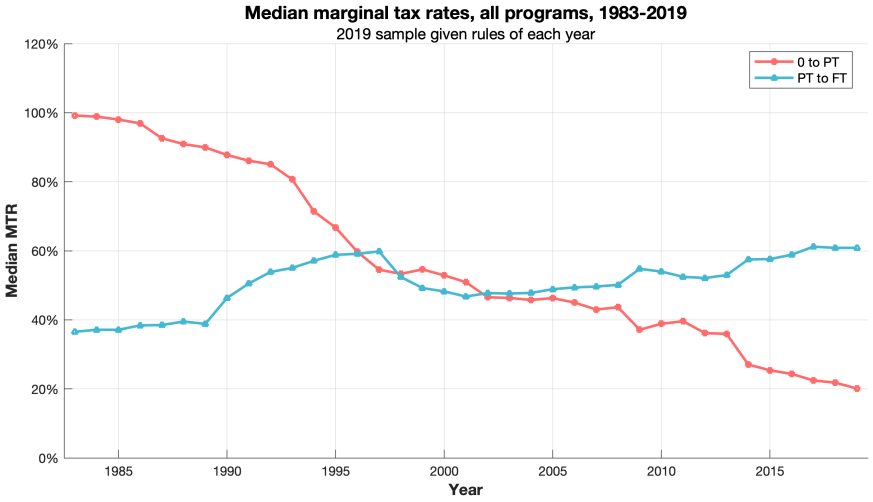
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- SNAP: major program which provides a benefits card for food purchases
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- Payroll and other state and federal income taxes

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- Payroll and other state and federal income taxes
- Using our calculators, can look at the effects of changes using the 2019 CPS; first across time, then for 2000 to 2019

MTRs with full program participation



Program participation (2019 sample)

Programs	Overall	0 hours	PT	FT
None	20.0 %	4.8 %	7.5 %	29.8 %
Med. (child, parent), SNAP	18.3 %	25.1 %	24.5 %	13.8 %
Med. (child, parent)	15.1 %	15.2 %	15.6 %	14.8 %
TANF, SNAP, Med. (child, parent)	3.5 %	8.9 %	4.3 %	1.3 %
All	3.3 %	2.1 %	2.1 %	4.2 %

Consumption (2019 sample)

Programs	No work		PT		FT	
	2000	2019	2000	2019	2000	2019
None	$ 1.6	$ 1.5	$ 16.1	$ 17.7	$ 24.7	$ 26.0
Med. (child, parent), SNAP	$ 11.2	$ 11.9	$ 21.7	$ 25.1	$ 28.3	$ 30.5
Med. (child, parent)	$ 7.7	$ 8.2	$ 20.3	$ 23.6	$ 28.1	$ 30.3
TANF, SNAP, Med. (child, parent)	$ 14.9	$ 14.9	$ 22.3	$ 25.7	$ 28.3	$ 30.6
All	$ 13.8	$ 14.9	$ 21.7	$ 27.6	$ 28.1	$ 31.8

Marginal tax rates (2019 sample)

Table: Marginal Tax Rates

Programs	0 to PT		PT to FT	
	2000	2019	2000	2019
None	-16.0 %	-29.0 %	27.1 %	29.5 %
Med. (child, parent), SNAP	16.7 %	-5.2 %	43.7 %	54.0 %
Med. (child, parent)	0.1 %	-22.8 %	32.7 %	42.2 %
TANF, SNAP, Med. (child, parent)	41.0 %	14.6 %	49.1 %	58.1 %
All	37.0 %	-1.2 %	45.6 %	64.2 %

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Overview

- Static model where single mothers choose discrete labor supply $h_i \in \{0, PT, FT\}$ and program participation $d_{ji} \in \{0, 1\}$ for programs $j \in J$
- Labor earnings depend on hours and an hours-specific wage, e.g. h_i, $w_i^{pt} \neq w_i^{ft}$

Overview

- Static model where single mothers choose discrete labor supply $h_i \in \{0, PT, FT\}$ and program participation $d_{ji} \in \{0, 1\}$ for programs $j \in J$
- Labor earnings depend on hours and an hours-specific wage, e.g. h_i, $w_i^{pt} \neq w_i^{ft}$
- Labor supply and program participation both come with a stochastic disutility cost, which will create incomplete takeup for benefits
  - The mean is allowed to shift over time, corresponding to major reforms
  - Assume full takeup for EITC and Housing, but simulate Housing offers since program is rationed (about 1 in 4 eligibles will receive an offer)
- The labor supply elasticities for each program are allowed to be different from each other and from private income, as the literature shows

Consumption

- $B_{ji}(h_i)$ are benefits for woman i from program j and depend on fixed characteristics (family size, state, year), labor earnings, and non-labor income N_i.
- Program labor supply effects may be heterogenous and different from private income, so benefits enter the utility function with a parameter γ_j

$$C^*(h_i, d_i) = N_i + w_i^{h_i} h_i - T_i + \gamma_{EITC} T_i^{EITC} + \sum_{j \in J} \gamma_j d_{ji} B_{ji}(h_i)$$

- C^* is scaled by $\omega_i = \sqrt{\frac{2}{1+k_i}}$ where k_i is number of kids (following Eckstein et al. 2019)

Utility

CRRA utility for $C^*(h_i, d_i)$, plus stochastic fixed costs α_i (part-time work), δ_i (full-time work), ψ_{ji} (program j):

$$V(h_i, d_i) = \frac{C^*(h_i, d_i)^{1-\eta} - 1}{1-\eta} - \alpha_i \mathbb{1}[h_i = PT] - \delta_i \mathbb{1}[h_i = FT] - \sum_{j \in J} d_{ji}\psi_{ji}$$

Where fixed costs follow a log normal distribution:

$$\log \alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2)$$
$$\log \delta_i \sim N(\mu_\delta, \sigma_\delta^2)$$
$$\log \psi_{ji} \sim N(\mu_{\psi_j}, \sigma_{\psi_j}^2), \ \forall j \in J$$

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Women observe shocks and choose $\max_{h_i, d_i} V(h_i, d_i)$

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CPS

- Sample of single mother heads-of-household from CPS restricted to:
  - Observations from 1983-2019
  - Non-labor income (NLI) below $1,000 monthly
  - Ages 20-44
  - Less than college degree education (but some college allowed)
- Fill in missing wages using Heckman-style polychotomous selection model with different wages at PT and FT (Lee 1983)
- We use the CPS sample to simulate the model, but largely avoid targeting moments from CPS statistics sample statistics

Benefits calculators

Benefits are calculated based on year, state, number of children, and total income.

Program	**Data Sources**
Taxes	NBER TaxSim, Tax Policy Foundation
EITC	TaxSim
SNAP	Mathematica Policy Reports, USDA
Housing	Fair Market Rent from Urban Institute's TRIM and HUD
TANF	TRIM
AFDC	TRIM, Congressional Research Service Reports
Medicaid (parents + adults)	Kaiser Family Foundation, TRIM, Macstats, CMS

Moments

Moment	**Estimate** (SE)	**Source**
Wage-participation elasticity	1 (-)	Meghir and Phillips [2010]
Wage-hours elasticity	.1 (.09)	Bishop et al. [2009]
Income-hours elasticity	-0.0065 (.007)	Bishop et al. [2009]
EITC elasticity	.026 (.003)	Schanzenbach and Strain [2021]
SNAP elasticity	-0.242 (.07)	Hoynes and Schanzenbach [2012]
Parent Medicaid elasticity	.015 (.008)	Kaestner et al. [2017]
AFDC elasticity (hours)	-16 (-)	Moffitt [1983]
TANF elasticity	0.077 (.009)	Bitler et al. [2006]
Housing	-0.036 (.016)	Jacob and Ludwig [2012]
Program takeup rates	rates	Administrative reports
PT labor supply	.177 (.002)	CPS
FT labor supply	.494 (.002)	CPS
Program takeup ratios	ratios	CPS

Identification

- Match 25 moments to calibrate 25 parameters
- Program elasticities $\rightarrow$ benefit effect γ_j
- Total income elasticity $\rightarrow$ CRRA curvature η
- Intensive and extensive wage elasticities $\rightarrow$ variance of PT and FT disutility σ_α^2, σ_δ^2
- PT and FT LS $\rightarrow$ mean of PT and FT disutility μ_α, μ_δ
- Program takeup rates $\rightarrow$ mean program disutilities μ_{ψ_j}
- Takeup ratio by family size $\rightarrow$ variance of program disutility $\sigma_{\psi_j}^2$

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Select moments

	Data	Estimate
% 0 LS	0.328	0.309
% PT LS	0.177	0.183
wage-participation elasticity	1.000	1.036
% FT LS	0.494	0.508
wage-hours elasticity	0.100	0.098
income-hours elasticity	-0.006	-0.006
EITC elasticity	0.106	0.105
TANF elasticity	0.077	0.077
AFDC elasticity	44.171	43.813
SNAP elasticity	0.236	0.237
Housing elasticity	0.037	0.037
Medicaid (parent) elasticity	0.015	0.015

Table: Model fit

Parameter estimates Full fit

Results

- Approach: fix the 2019 sample (most policy relevant), and examine differences in outcomes when facing rules from 2000
- Use the model to simulate choices, MTRs, consumption, and utility in the two policy regimes
- Break down role of each program

Outcomes under 2000 and 2019 rules

Moment	2000 rules 2000 choices	2019 rules 2000 choices	2019 rules 2019 choices	Change, 2000 to 2019
% 0 hours	0.24	-	0.19	-20.06 %
% PT hours	0.25	-	0.31	20.54 %
% FT hours	0.50	-	0.50	-0.65 %
TANF participation	0.12	-	0.08	-33.65 %
SNAP participation	0.39	-	0.40	4.45 %
Medkid participation	0.70	-	0.70	-0.14 %
Medpar participation	0.24	-	0.34	41.19 %
Mean consumption ($100)	$ 23.10	$ 24.84	$ 25.98	12.45 %
Mean utility	2.60	2.72	2.76	6.38 %
Net expenditure ($100K)	$ 61.81	$ 78.82	$ 86.96	40.69 %
Atkinson (consumption)	0.047	0.052	0.051	8.00 %
Atkinson (utility)	0.033	0.032	0.032	-3.55 %

Utility under 2000 vs. 2019 rules

- Consumption is up, utility up by a smaller amount
  - Women consume more because they are working more – but pay disutility cost
- Inequality in consumption is up, as seen by Atkinson measure or the 90-10 percentile ratio figure
  - However, inequality in utility is somewhat lower, reflecting the increased LS disutility by those enjoying more consumption
- Consumption equivalents: how much would you have to give 2000 agent to make them equally as well off as under the 2019 rules?

Consumption equivalents

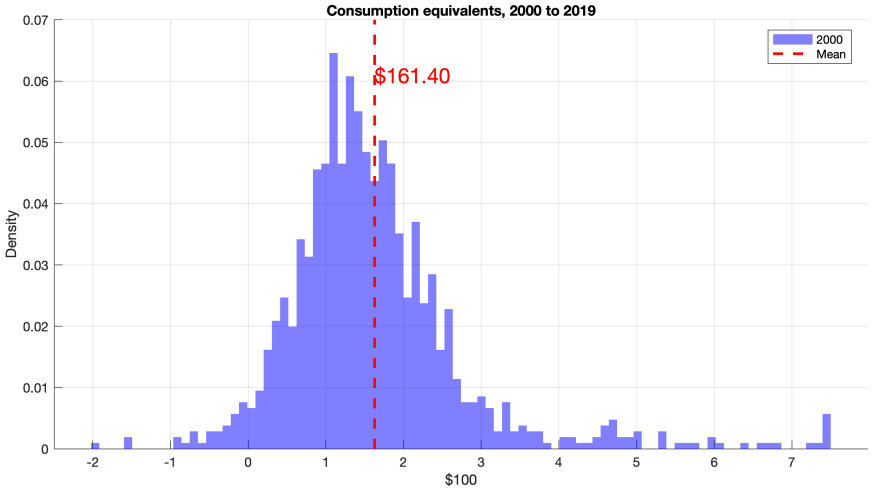


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Revenue-neutral reform

- Net government expenditures for our sample went up about 40% from the 2000 rules to 2019 rules
  - Can use the model to explore alternative, revenue-neutral reforms
- Begin simple: what if, instead of decreasing MTRs (increasing benefits where $H > 0$), we had increased baseline guaranteed benefits?

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  - Can use the model to explore alternative, revenue-neutral reforms
- Begin simple: what if, instead of decreasing MTRs (increasing benefits where $H > 0$), we had increased baseline guaranteed benefits?
- We find an increase to the baseline guarantee of SNAP, TANF, Medicaid, and housing of **17%** would have led to the same rise in net expenditures, but with opposite employment effects and smaller welfare gains:
  - The employment rate goes down by 7-percentage-points (9%)
  - Mean consumption increases by 3.3% (down from 12%)
  - Utility increases by 3.1% (down from 6.4%)

Outcomes under revenue neutral reform

Moment	2000 rules	2019 rules	Change	Alt.	Change
% 0 hours	0.24	0.19	-20.1 %	0.31	28.8 %
% PT hours	0.25	0.31	20.5 %	0.24	-5.4 %
% FT hours	0.50	0.50	-0.7 %	0.45	-11.2 %
Mean utility	2.72	2.76	6.4 %	2.68	3.1 %
Mean consumption ($100)	$ 24.84	$ 25.66	11.0 %	$ 23.98	3.8 %
Total private income ($100K)	$ 182.74	$ 186.62	0.0 %	$ 166.32	-9.0 %
Atkinson (consumption)	0.052	0.051	8.0 %	0.055	15.9 %
Atkinson (utility)	0.032	0.032	-3.5 %	0.027	-16.7 %

Subgroup impacts

- Under the pivot, increased benefits encouraged labor supply, so consumption went up by more than the increase in expenditures; opposite as under the alternative
- But there may be distributional differences

Subgroup impacts

- Under the pivot, increased benefits encouraged labor supply, so consumption went up by more than the increase in expenditures; opposite as under the alternative
- But there may be distributional differences
- Classify sample as 0 hours, part-time, and full-time workers based on simulated work probabilities, and look at outcomes conditional on being in each work status:
  - Expenditures under the alternative reform mostly go towards non-workers, who do significantly better than before
  - Part-time workers see moderate (but smaller) consumption and utility gains; impact on full-time workers small

Changes from 2000 baseline

Moment	Non-workers		Part-time		Full-time	
	2019	Alt.	2019	Alt.	2019	Alt.
Mean consumption ($100)	4.0 %	15.4 %	15.4 %	8.2 %	6.5 %	2.3 %
Mean utility	1.6 %	7.8 %	7.9 %	4.9 %	4.5 %	2.8 %
Net Expenditure	6.9 %	19.4 %	42.0 %	20.7 %	-1011.1 %	-316.1 %
Total Consumption	5.5 %	15.5 %	17.0 %	8.4 %	7.2 %	2.2 %

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Next steps

- Extending the revenue-neutral reform exercise to examine some frequently discussed policy proposals
- Considering other counterfactuals, as well as extending analysis further back (e.g. looking at the AFDC to TANF transition)
- Additional features to add to the model include correlation between LS shocks and program shocks as well as private health insurance

Conclusion

- The U.S. social safety net underwent the **Great MTR Pivot** from 1983-2019; MTRs fell sharply at the lower end of earnings distribution, increased at higher levels
- Estimate a literature-consistent model of LS and program participation for single mothers
- Use the model to look at the evolution from 2000 to 2019 and show:
  - Employment rate increased by 5pp, mostly through PT work
  - Monthly consumption rose by $290 (11%) on average; utility by less (about 3.1%)
  - Inequality in consumption (but not utility) increased
- Primary driver in changes is the expanded EITC, with other programs playing smaller roles
- Revenue-neutral reform suggests that increasing baseline benefits while leaving phaseouts and other parameters untouched would have been sub-optimal

Thank you

Sample statistics return

Variable	Mean (SD)
% White non-hispanic	44.4
Number of children	1.94 (.97)
% part-time	17.68
Part-time wage	$11.09 (0.055)
% full-time	50.06
Full-time wage	$13.48 (0.034)
% less than HS	31.18
% HS	32.43
% more than HS	36.39
N	67,550

Moments return

Takeup rates are collected from administrative reports.

Program	Period	Estimate	Source
AFDC	83-96	.810	HHS
TANF	97-19	.389	
SNAP	83-93	.785	HHS
SNAP	94-04	.866	
SNAP	05-19	.820	
Medicaid (child)	83-97	.728	HHS, AHRQ
Medicaid (child)	98-19	.868	
Medicaid (parent)	83-02	.796	
Medicaid (parent)	03-12	.786	
Medicaid (parent)	14-19	.769	

Table: Takeup rates

Takeup ratios

return

Takeup ratios are calculated from the CPS; family size gives ratio of takeup in single- to multi-child families and hours gives ratio of takeup among non-workers to that of workers.

Program	**Family size**	**Hours**
AFDC	0.98 (.011)	1.2 (.044)
TANF	.91 (.039)	2.22 (.128)
SNAP	.98 (.007)	1.45 (.013)
Medicaid (child)	1.02 (.006)	1.1 (.009)
Medicaid (parent)	.86 (.007)	1.67 (.012)

Table: Takeup ratios

Select parameter estimates

return

	Estimate	SE
Mean PT disutility ($\mathbb{E}[\alpha_i]$)	1.486	0.034
Mean FT disutility ($\mathbb{E}[\delta_i]$)	1.551	0.032
CRRA curvature (η)	0.720	0.007
EITC coef. (γ_{EITC})	1.103	0.022
TANF coef. (γ_{TANF})	0.278	0.008
Mean TANF disutility ($\mathbb{E}[\psi_{TANF,i}]$)	0.112	0.004
AFDC coef. (γ_{AFDC})	0.285	0.007
Mean AFDC disutility ($\mathbb{E}[\psi_{AFDC,i}]$)	0.485	0.068
SNAP coef. (γ_{SNAP})	0.463	0.011
Mean SNAP disutility, 94-04 ($\mathbb{E}[\psi_{SNAP,i}\|1994-2004]$)	0.037	0.001
Mean SNAP disutility, 2005-2019 ($\mathbb{E}[\psi_{SNAP,i}\|1983-1993]$)	0.033	0.001
Housing coef. (γ_{house})	0.084	0.003
Mean Medicaid (child) disutility, 98-19 ($\mathbb{E}[\psi_{SNAP,i}\|1998-2019]$)	0.072	0.002
SNAP coef. (γ_{medpar})	0.497	0.011
Mean Medicaid (parent) disutility, 03-13 ($\mathbb{E}[\psi_{medpar,i}\|2003-2013]$)	3.334	0.157

Table: Selected parameter Estimates

Full parameter estimates pt. 1 return

	Estimate	SE
Mean PT disutility ($\mathbb{E}[\alpha_i]$)	1.486	0.034
S.D. PT disutility (σ_α)	0.501	0.026
Mean FT disutility ($\mathbb{E}[\delta_i]$)	1.551	0.032
S.D. FT disutility (σ_δ)	0.295	0.004
CRRA curvature (η)	0.720	0.007
EITC coef. (γ_{EITC})	1.103	0.022
TANF coef. (γ_{TANF})	0.278	0.008
Mean TANF disutility ($\mathbb{E}[\psi_{TANF,i}]$)	0.112	0.004
S.D. TANF disutility ($\sigma_{\psi_{TANF}}$)	0.876	-
AFDC coef. (γ_{AFDC})	0.285	0.007
Mean AFDC disutility ($\mathbb{E}[\psi_{AFDC,i}]$)	0.485	0.068
S.D. AFDC disutility ($\sigma_{\psi_{AFDC}}$)	3.000	-
SNAP coef. (γ_{SNAP})	0.463	0.011

Table: Parameter Estimates

Full parameter estimates pt. 2 return

	Estimate	SE
Mean SNAP disutility, 83-93 ($\mathbb{E}[\psi_{SNAP,i}\|1983-1993]$)	0.068	0.002
S.D. SNAP disutility ($\sigma_{\psi_{SNAP}}$)	0.008	-
Mean SNAP disutility, 94-04 ($\mathbb{E}[\psi_{SNAP,i}\|1994-2004]$)	0.037	0.001
Mean SNAP disutility, 2005-2019 ($\mathbb{E}[\psi_{SNAP,i}\|1983-1993]$)	0.033	0.001
Housing coef. (γ_{house})	0.084	0.003
Mean Medicaid (child) disutility, 83-97 ($\mathbb{E}[\psi_{SNAP,i}\|1983-1997]$)	0.092	0.003
S.D. Medicaid (child) disutility ($\sigma_{\psi_{medkid}}$)	0.011	-
Mean Medicaid (child) disutility, 98-19 ($\mathbb{E}[\psi_{SNAP,i}\|1998-2019]$)	0.072	0.002
SNAP coef. (γ_{medpar})	0.497	0.011
Mean Medicaid (parent) disutility, 83-02 ($\mathbb{E}[\psi_{medpar,i}\|1983-2002]$)	1.977	0.099
S.D. Medicaid (parent) disutility ($\sigma_{\psi_{SNAP}}$)	3.000	-
Mean Medicaid (parent) disutility, 03-13 ($\mathbb{E}[\psi_{medpar,i}\|2003-2013]$)	3.334	0.157
Mean Medicaid (parent) disutility, 2014-2019 ($\mathbb{E}[\psi_{medpar,i}\|2014-1919]$)	1.568	0.087

Table: Parameter Estimates

Model fit pt. 1 return

	Data	Estimate
% 0 LS	0.328	0.309
% PT LS	0.177	0.183
wage-participation elasticity	1.000	1.036
% FT LS	0.494	0.508
wage-hours elasticity	0.100	0.098
income-hours elasticity	-0.006	-0.006
EITC elasticity	0.106	0.105
TANF elasticity	0.077	0.077
TANF takeup	0.389	0.390
TANF ratio	4.253	4.254
AFDC elasticity	44.171	43.813
AFDC takeup	0.810	0.852
AFDC ratio	1.521	1.562
SNAP elasticity	0.236	0.237

Table: Model fit

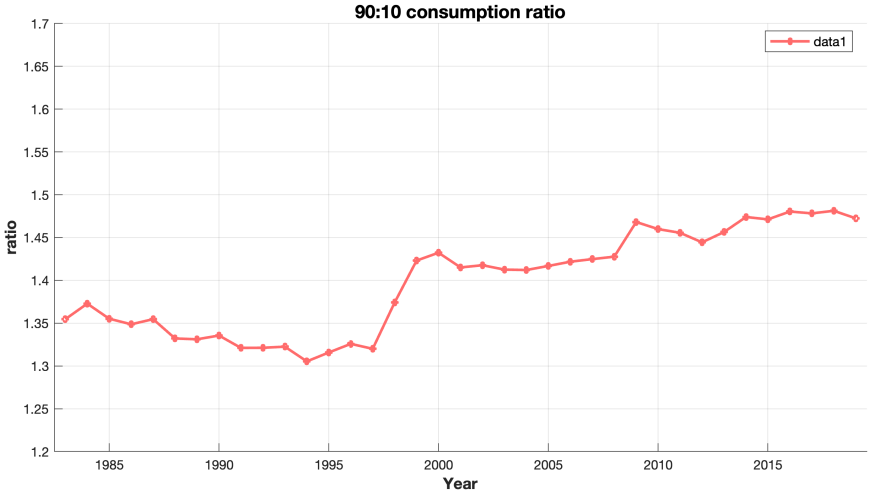
Model fit pt. 2 return

	Data	Estimate
SNAP takeup, 83-93	0.785	0.769
SNAP ratio	1.640	1.577
SNAP takeup, 94-04	0.845	0.817
SNAP takeup, 05-19	0.895	0.857
Housing elasticity	0.037	0.037
Medicaid (child) takeup, 83-97	0.772	0.762
Medicaid (child) ratio	1.720	1.613
Medicaid (child) takeup, 98-19	0.826	0.741
Medicaid (parent) elasticity	0.015	0.015
Medicaid (parent) takeup, 83-02	0.794	0.788
Medicaid (parent) ratio	1.126	1.157
Medicaid (parent) takeup, 03-13	0.706	0.711
Medicaid (parent) takeup, 14-19	0.773	0.781

Table: Model fit

90th to 10th percentile consumption

return



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